

## Ø 5 mm Film Dielectric Trimmers



### FEATURES

- Housing diameter 5 mm
- Top and bottom or top adjustment
- Round head
- Mounting: radial
- Material categorization: for definitions of compliance please see [www.vishay.com/doc?99912](http://www.vishay.com/doc?99912)

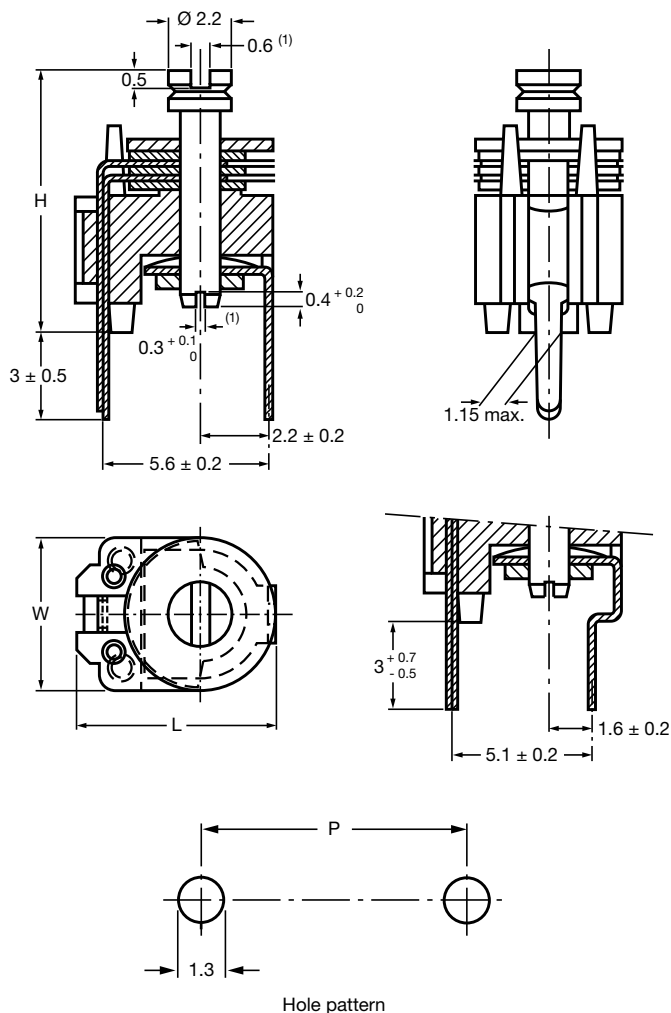

**RoHS**  
COMPLIANT

### APPLICATIONS

- Impedance matching circuits
- RF
- Medical
- For consumer and industrial equipment

### QUICK REFERENCE DATA

|                                                            |                           |                                                                                                                                                                                                                                                                                                    |
|------------------------------------------------------------|---------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Rated DC voltage                                           |                           | 150 V <sub>DC</sub>                                                                                                                                                                                                                                                                                |
| Test DC voltage for 1 min                                  |                           | 300 V <sub>DC</sub>                                                                                                                                                                                                                                                                                |
| Allowed maximum AC voltage                                 |                           | 15 V <sub>AC</sub>                                                                                                                                                                                                                                                                                 |
| Maximum contact resistance                                 |                           | 10 mΩ                                                                                                                                                                                                                                                                                              |
| Minimum insulation resistance                              |                           | 10 000 MΩ                                                                                                                                                                                                                                                                                          |
| Category temperature range                                 | PP                        | -40 °C to +70 °C                                                                                                                                                                                                                                                                                   |
|                                                            | PTFE                      | -40 °C to +85 °C                                                                                                                                                                                                                                                                                   |
| Climatic category (IEC 60068)                              | PP                        | 40/070/21                                                                                                                                                                                                                                                                                          |
|                                                            | PTFE                      | 40/085/21                                                                                                                                                                                                                                                                                          |
| Minimum storage temperature                                |                           | -55 °C                                                                                                                                                                                                                                                                                             |
| Related specification                                      |                           | IEC 60418-1 and 4                                                                                                                                                                                                                                                                                  |
| Effective angle of rotation                                |                           | 180° (rotation in 180° only, see "Life of Trimmer")                                                                                                                                                                                                                                                |
| Operating torque                                           | C <sub>max.</sub> < 20 pF | 1 mNm to 15 mNm                                                                                                                                                                                                                                                                                    |
|                                                            | C <sub>max.</sub> ≥ 20 pF | 1 mNm to 25 mNm                                                                                                                                                                                                                                                                                    |
| Maximum axial thrust                                       |                           | 2 N                                                                                                                                                                                                                                                                                                |
| Capacitance range (C <sub>min.</sub> / C <sub>max.</sub> ) |                           | 0.35 pF / 1.5 pF to 4 pF / 27 pF                                                                                                                                                                                                                                                                   |
| Life of trimmer                                            |                           | Maximum 10 cycles: rotation in 180° only (the electrical and mechanical performance is not guaranteed if rotated beyond 10 cycles)                                                                                                                                                                 |
| Quality level                                              |                           | Sampling and data evaluation for quality level in accordance with "MIL-STD-105D" and "IEC 60410":<br><div> <div>&lt; 0.15 % major defects</div> <div>&lt; 0.65 % minor defects</div> </div> Each capacitor is tested for minimum C <sub>max.</sub> and is also subjected to the full test voltage. |

**DIMENSIONS** in millimeters


Trimmers BFC2 808 ..... series, with round head

| CAPACITANCE AND RELEVANT PHYSICAL DIMENSIONS |                     |                     |                     |
|----------------------------------------------|---------------------|---------------------|---------------------|
| $C_{\min.} / C_{\max.}$<br>(pF)              | $H_{\max.}$<br>(mm) | $W_{\max.}$<br>(mm) | $L_{\max.}$<br>(mm) |
| 0.35 / 1.5                                   | 7.0                 | 5.5                 | 7.3                 |
| 1.5 / 5                                      | 7.0                 | 5.5                 | 7.3                 |
| 3 / 10                                       | 7.0                 | 5.5                 | 7.3                 |
| 3 / 15                                       | 8.8                 | 5.5                 | 7.3                 |
| 4 / 20                                       | 8.8                 | 5.5                 | 7.3                 |
| 4 / 27                                       | 9.0                 | 6.2                 | 7.8                 |

**MOUNTING**

The trimmer has a lead pitch of 5.08 mm or 5.6 mm and can be mounted on printed-circuit boards with a minimum hole diameter of 1.25 mm.

**PACKAGING**

Bulk packaged in cardboard boxes lined with expanded plastic, 1000 units per box.

**ORDERING INFORMATION**

| $C_{min.} / C_{max.}$<br>(pF) | CATALOG NUMBER BFC2 808 .....             |                                     |                                      |
|-------------------------------|-------------------------------------------|-------------------------------------|--------------------------------------|
|                               | TOP AND BOTTOM ADJUSTMENT<br>(P = 5.6 mm) | TOP ADJUSTMENT ONLY<br>(P = 5.6 mm) | TOP ADJUSTMENT ONLY<br>(P = 5.08 mm) |
| <b>POLYTETRAFLUORETHYLENE</b> |                                           |                                     |                                      |
| 0.35 / 1.5                    | 22158                                     | -                                   | -                                    |
| <b>POLYPROPYLENE</b>          |                                           |                                     |                                      |
| 1.2 / 5                       | -                                         | 24508                               | -                                    |
| 1.5 / 5                       | 23508                                     | -                                   | 20508                                |
| 1.5 / 7                       | -                                         | 24708                               | -                                    |
| 3 / 10                        | 23109                                     | -                                   | 20109                                |
| 3 / 15                        | 23159                                     | -                                   | 20159                                |
| 4 / 20                        | 23209                                     | -                                   | 20209                                |
| 4 / 27                        | 23279                                     | -                                   | 20279                                |

**ELECTRICAL DATA**

| GUARANTEED<br>MAX. $C_{min.}$ /<br>MIN. $C_{max.}$<br>AT 200 kHz<br>(pF) | $\tan \delta$<br>AT $C_{max.} \times 10^{-4}$ |           | TEMP.<br>COEFF. <sup>(1)</sup><br>( $10^{-6}/K$ ) | MIN. $f_{res}$<br>AT $C_{max.}$<br>(MHz) | COLOR<br>OF<br>BASE | SMALLEST<br>PACKAGING<br>QUANTITY | CATALOG NUMBER<br>BFC2 ... ..    |
|--------------------------------------------------------------------------|-----------------------------------------------|-----------|---------------------------------------------------|------------------------------------------|---------------------|-----------------------------------|----------------------------------|
|                                                                          | 1 MHz                                         | 100 MHz   |                                                   |                                          |                     |                                   |                                  |
| 0.35 / 1.5                                                               | $\leq 10$                                     | -         | $-450 \pm 550$                                    | -                                        | -                   | 1000                              | .... 808 22158                   |
| 1.2 / 5                                                                  | $\leq 10$                                     | -         | $-200 \pm 550$                                    | -                                        | Grey                | 1000                              | .... 808 24508                   |
| 1.5 / 5                                                                  | $\leq 10$                                     | $\leq 25$ | $-200 \pm 550$                                    | 700                                      | Grey                | 1000                              | .... 808 20508<br>.... 808 23508 |
| 1.5 / 7                                                                  | $\leq 10$                                     | -         | $-50 \pm 550$                                     | -                                        | Grey                | 1000                              | .... 808 24708                   |
| 3 / 10                                                                   | $\leq 10$                                     | $\leq 25$ | $-250 \pm 550$                                    | 500                                      | Yellow              | 1000                              | .... 808 20109<br>.... 808 23109 |
| 3 / 15                                                                   | $\leq 10$                                     | $\leq 25$ | $-250 \pm 550$                                    | 400                                      | Blue                | 1000                              | .... 808 20159<br>.... 808 23159 |
| 4 / 20                                                                   | $\leq 10$                                     | $\leq 25$ | $-250 \pm 400$                                    | 300                                      | Green               | 1000                              | .... 808 20209<br>.... 808 23209 |
| 4 / 27                                                                   | $\leq 10$                                     | $\leq 25$ | $-250 \pm 400$                                    | 300                                      | Red                 | 1000                              | .... 808 20279<br>.... 808 23279 |

**Note**

<sup>(1)</sup> C: 60 % to 80 % of  $C_{max.}$ ;  $T_{amb.}$ : from +20 °C to +70 °C

**SOLDERING CONDITIONS**

For general soldering conditions and wave soldering profile, we refer to the application note "Soldering Guidelines for Film Capacitors": [www.vishay.com/doc?28171](http://www.vishay.com/doc?28171)

**TEST PROCEDURES AND REQUIREMENTS**

| IEC<br>60418-1<br>CLAUSE | IEC 60068<br>TEST<br>METHOD | TEST                        | PROCEDURE            | REQUIREMENTS                                                                                      |
|--------------------------|-----------------------------|-----------------------------|----------------------|---------------------------------------------------------------------------------------------------|
| 4.2                      |                             | Method of mounting          | Method A             |                                                                                                   |
| 14                       |                             | Capacitance drift           | After TC measurement | $\Delta C/C: \leq 3\%$ for $C_{max.} \leq 10$ pF<br>$\Delta C/C: \leq 2\%$ for $C_{max.} > 10$ pF |
| 19                       |                             | Thrust                      | Axial thrust of 2 N  | $\Delta C/C: \leq 0.4\%$                                                                          |
| 21                       |                             | Robustness of terminations: |                      |                                                                                                   |
| 21.1                     | Ua                          | Tensile                     | 1 N                  | No damage                                                                                         |
| 21.2                     | Ub                          | Bending                     | 1 cycle              | No damage                                                                                         |

**TEST PROCEDURES AND REQUIREMENTS**

| IEC 60418-1<br>CLAUSE | IEC 60068<br>TEST<br>METHOD | TEST                                       | PROCEDURE                                                                                                                                            | REQUIREMENTS                                                                                                                                                                                                                                                                                                                                                                                                                                           |
|-----------------------|-----------------------------|--------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 22                    | Na                          | Rapid change of temperature                | 1 cycle; 0.5 h at lower and 0.5 h at upper category temperature                                                                                      | $\Delta C/C: \leq 2.5 \%$                                                                                                                                                                                                                                                                                                                                                                                                                              |
| 23                    | T                           | Soldering:                                 |                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
|                       | Ta                          | Solderability                              | Solder bath immersion 3 mm; 235 °C; 2 s                                                                                                              | Good wetting;<br>no mechanical damage                                                                                                                                                                                                                                                                                                                                                                                                                  |
|                       | Tb                          | Resistance to heat                         | Solder bath: 260 °C; 10 s                                                                                                                            | No mechanical damage                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| 24                    | Eb                          | Impact bump                                | 4000 $\pm$ 10 bumps; 40 g; 6 ms                                                                                                                      | $\Delta C/C: \leq 1 \%$ ;<br>no mechanical damage                                                                                                                                                                                                                                                                                                                                                                                                      |
| 25                    | Fc                          | Vibration                                  | Frequency 10 Hz to 55 Hz;<br>amplitude 0.75 mm;<br>1.5 h                                                                                             | $\Delta C/C: \leq 1 \%$ ;<br>no mechanical damage                                                                                                                                                                                                                                                                                                                                                                                                      |
| 26                    |                             | Climatic sequence:                         |                                                                                                                                                      | $\Delta C/C: \leq 4 \%$                                                                                                                                                                                                                                                                                                                                                                                                                                |
| 26.1                  | B                           | Dry heat                                   | 16 h at upper category temperature                                                                                                                   | $\tan \delta$ or PP and PTFE foil: $\leq 15 \times 10^{-4}$<br>$\tan \delta$ for PC foil: $\leq 80 \times 10^{-4}$<br><br>$R_{ins.}: \geq 10\,000\,M\Omega$<br>Rotor contact R: $\leq 10\,m\Omega$                                                                                                                                                                                                                                                     |
| 26.2                  | D                           | Damp heat accelerated,<br>first cycle      | 1 cycle; 24 h; +40 °C;<br>95 % to 100 % RH                                                                                                           | Voltage proof:<br>300 V for 1 min                                                                                                                                                                                                                                                                                                                                                                                                                      |
| 26.3                  | Aa                          | Cold                                       | 16 h; -40 °C                                                                                                                                         | Visual examination:<br>no mechanical damage                                                                                                                                                                                                                                                                                                                                                                                                            |
| 26.5                  |                             | Damp heat accelerated,<br>remaining cycles | 1 cycle; 24 h; +40 °C;<br>95 % to 100 % RH                                                                                                           | Operating torque:<br>1 mNm to 20 mNm for $C_{max.} < 20\,pF$ ;<br>1 mNm to 30 mNm for $C_{max.} \geq 20\,pF$                                                                                                                                                                                                                                                                                                                                           |
| 27                    | Ca                          | Damp heat steady state                     | 21 days; +40 °C;<br>90 % to 95 % RH                                                                                                                  | $\Delta C/C: \leq 3 \%$<br><br>$\tan \delta$ for PP and PTFE foil: $\leq 15 \times 10^{-4}$ ;<br>$\tan \delta$ for PC foil: $\leq 80 \times 10^{-4}$<br><br>$R_{ins.}: \geq 10\,000\,M\Omega$ ;<br>Rotor contact R: $\leq 10\,m\Omega$<br><br>Voltage proof:<br>300 V for 1 min<br><br>Visual examination:<br>no mechanical damage<br><br>Operating torque:<br>1 mNm to 20 mNm for $C_{max.} < 20\,pF$ ;<br>1 mNm to 30 mNm for $C_{max.} \geq 20\,pF$ |
| 29                    |                             | Mechanical endurance                       | 10 cycles<br><br>Maximum 10 cycles: rotation in 180° only. (The electrical and mechanical performance is not guaranteed if rotated beyond 10 cycles) | $\Delta C/C: \leq 3 \%$<br><br>$\Delta C/C$ after axial thrust: $\leq 0.3 \%$ ;<br>rotor contact R: $\leq 10\,m\Omega$<br><br>Voltage proof:<br>300 V for 1 min<br><br>Visual examination:<br>no mechanical damage<br><br>Operating torque:<br>0.5 mNm to 22.5 mNm for $C_{max.} < 20\,pF$ ;<br>0.5 mNm to 30 mNm for $C_{max.} \geq 20\,pF$                                                                                                           |



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